

Ship/HULL NO.	Ship/HULL NAME	JON:	ICN:	Rev:	Key Op:
YON-305	FLCY SASEBO FUEL DEPT		89NSC85001R		A01

TITLE:

SA YON-0120D for Cargo Pump; accomplish

JOB SPECIFICATION

Anticipated Award Date: 7/1/2026	Production Completion Date: 11/18/2026	PREPARING ACTIVITY:	
POP Start Date: 7/22/2026	POP End Date: 12/31/2026	SASEBO	
PREPARED BY: K.SAITO	CODE M_SK	DATE 5/19/2025	
PATAPPROVED BY: S.MURAKAMI	CODE S225		
CONCURRED BY: N.SASHI	CODE S232		
ASSIST PLANNER'S NAME: N.NAGATO	CODE H_NA		
ASSIST PLANNER'S NAME: S.TERASHITA	CODE P_TE		
ASSIST PLANNER'S NAME:	CODE		
DESCRIPTION:			
JSN(S): DFSP-0003		CASREP #	
S/F CONTACT: REYES DONNY CIV / SANTOS FRANK		CWP #	CCNS #
Approved by S300/S900 <i>[Signature]</i> Date: 5/20/2025		AVL: 6 CM	5GM
		LWC:	
Work Completed			
S/Force:		Rate:	Date:
Ship Supt:		Date:	SRF CONTROL NO.: 1301362787
WORK CERTIFICATION SIGNATURES			
COMPLETION OF WORK REVIEW:	BADGE NO.:	SHOP/CODE:	PHONE:
			DATE:

Ship/HULL NO.	Ship/HULL NAME	JON:	ICN:	Rev:	Key Op:
YON-305	FLCY SASEBO FUEL DEPT		89NSC85004 ¹¹ R		A01

TITLE:

SA YON-0120D for Cargo Pump; accomplish

1. SCOPE:

- 1.1 Title: ShipAlt YON-0120D for Cargo Pump; accomplish
- 1.2 Location of Work:
 - 1.2.1 Pump Room
- 1.3 Identification:
 - 1.3.1 Listed in 2.2 and 2.3 for Modification of Cargo Pump, Foundation, and Piping located in 1.2.1

2. REFERENCES:

- 2.1 Standard Items ^{FY26}~~(FY25)~~
- 2.2 185-8970193, S/A YON-0120D YON 305, Foundation Modifications Incidental Cargo Pump Install
- 2.3 185-8970194, S/A YON-0120D YON 305, Cargo Pump Installation

3. REQUIREMENTS:

3.1 Clean bilge areas and piping affected by this work item to a gas/oil free condition, in accordance with 29 CFR 1915 Subpart B.

(V) "REPORT"

3.1.1 Submit one legible copy of Marine Chemist certificate to FLCY SASEBO FUEL DEPT including certified location.

3.2 Provide each portable air driven bilge pump, high pressure pump with each necessary nozzle, suction hose, and required equipment to high pressure clean and gas-free each bilge space located in 1.2.1.

3.2.1 Ventilate, empty, clean, and maintain each bilge area throughout the work item performance period.

3.2.1.1 Ensure that each harmful vapor, fume, and mist is ventilated to the exterior of the vessel.

3.2.1.2 Remove and dispose of waste liquid that is not being stored for reuse to a Contractor furnished waste liquid conveyance.

3.2.1.3 Provide watches and following spill kits (absorbent pads, booms, etc. for oil or chemical spill containment) at each affected areas where waste liquid is being transferred to ensure that each liquid is not released to any shipboard compartment or the sea. In case of a spill, take immediate action to contain the spill.

Ship/HULL NO.	Ship/HULL NAME	JON:	ICN:	Rev:	Key Op:
YON-305	FLCY SASEBO FUEL DEPT		11 89NSC85004R		

TITLE:

SA YON-0120D for Cargo Pump; accomplish

3.2.1.4 Identify each transfer hose to the conveyance by affixing the company's name and space location pumping from on each hose.

3.2.1.5 Establish and maintain communication between the pumping station and the conveyance when the waste liquid is being transferred.

(V) "INSPECT PIPING AND HOSE ALIGNMENT"

3.2.1.6 Inspect alignment of each hose, valve and piping in the flow path prior to the start of pumping, purging, or flushing.

3.2.1.7 Accomplish a periodic walkthrough of the piping, hose, and valve alignments for leaks at least once an hour.

3.2.1.8 Pumping after sunset is not authorized unless approved by FLCY SASEBO FUEL DEPT.

3.3 Accomplish the requirements of 2.2 and 2.3 for modification of each cargo pump, foundation, and piping listed in 1.3.1 including each general note.

3.3.1 Prior to installation of each foundation, verify each dimension to adjust and determine the pump foundation installation position and fastening position, using Note 10 of 2.2, Note 20 of 2.3, ENCLOSURE (B), ENCLOSURE (C), and ENCLOSURE (D) for guidance.

3.3.2 Accomplish the requirements of **009-12** of 2.1, including Table One, Column A, Line One through 10 for welding.

3.3.2.1 Fabricate each backing ring in accordance with Line 6.

3.3.3 Accomplish the requirements of **009-71** of 2.1 for shop hydrostatic test of fabricated piping except each new relief valve (V-1 of 2.3), using fresh water at 113 (+3/-0) PSIG.

3.3.4 Accomplish the requirements of **009-12** of 2.1, including Table 4, Column A, Line One through 7 for welding.

(V) (G) "SHAFT ALIGNMENT"

3.3.5 Align each shaft between the pump and vertical drive in accordance with Note 20 of 2.3. **-FLCY SASEBO, S130**

3.3.5.1 Install and fit new shims conforming to ASTM A 240 to accomplish alignment.

3.3.5.2 Drill and ream each foundation. Fit and install each new dowel pin.

3/15

Ship/HULL NO.	Ship/HULL NAME	JON:	ICN: 11 89NSC85001R	Rev:	Key Op: A01
YON-305	FLCY SASEBO FUEL DEPT				

TITLE:

SA YON-0120D for Cargo Pump; accomplish

3.3.6 Align the piping to the pump. Piping must be supported independently and must not impose a strain on the pump. Accomplishment of alignment by adjustment of the first adjacent hanger is to be considered within the scope of the work.

(V) (G) "OPERATING PRESSURE TEST"

3.4 Witness an operating pressure test to be conducted by Ship's Force to verify tightness of each mechanical joint affected by this work item, until the specified criteria is verified. Allowable leakage: None. **-FLCY SASEBO, S130**

(V) (G) "OPERATIONAL TEST"

3.5 Accomplish an operational test of cargo pump listed in 1.3.1. Record the data in accordance with ENCLOSURE (A). **-FLCY SASEBO, S130**

3.6 Accomplish the requirements of **009-32** of 2.1 for each new and disturbed surface.

4. NOTES:

4.1 Initial Condition - Request replacement of cargo pump and associated components IAW SHIPALT YON-0120D.

4.2 SRF Tool Room will issue each pressure gage, dial indicator, and torque wrench when required.

4.3 Contractor is required to provide material not specifically listed as GFM.

4.4 Attachment A is provided from S246 as response to TP request.

4.5 The cleaning and gas free work of each cargo fuel oil tank (S-1, P-1, S-2, P-2, S-3 and P-3) will be covered by ICN: 38NSC85002_A01.

4.6 The floating shaft has been ordered with the estimated length described in Note 20 of 2.3.

5. GOVERNMENT FURNISHED MATERIAL (GFM) :

5.1 SRF will provide following materials.

MATERIAL LIST									
ITEM NO.	PC NO.	REF NO.	M / C	DESCRIPTION	STOCK NUMBER	HAZ MAT	DOC NO.	QTY REQD/ UNIT	PARA NO.
501	1	1	M	ANGLE,ANGLE, 5 X 3-1/2 X 3/8 MTL SPEC: ASTM A36 MATERIAL: OSS	9B 4730	9		6/LF	3.3

4/15

Ship/HULL NO.	Ship/HULL NAME	JON:	ICN: 11 89NSC85004R	Rev:	Key Op: A01
YON-305	FLCY SASEBO FUEL DEPT				

TITLE:

SA YON-0120D for Cargo Pump; accomplish

MATERIAL LIST									
ITEM NO.	PC NO.	REF NO.	M / C	DESCRIPTION	STOCK NUMBER	HAZ MAT	DOC NO.	QTY REQD/ UNIT	PARA NO.
502	2	1	M	CHANNEL, CHANNEL, 15" X 33.9 MTL SPEC: ASTM A36 MATERIAL: OSS	9B 4730	9		10/LF	3.3
503	3	1	M	BOLT HEX HD, BOLT, HEX HD, 3/4-10UNC-2A - 1-1/2 LG MIL SPEC: ASME B.18.2.1 MATERIAL: ZINC PLATED	9B 5330	9		8/EA	3.3
504	4	1	M	NUT, SLF LKG, NUT, SLF LKG 3/4- 10UNC-3B MIL SPEC: NASM 25027 MATERIAL: ZINC PLATED	9B 5330	9		8/EA	3.3
505	5	1	M	WASHER, FLAT, WASHER, FLAT, 3/4" DIA MIL SPEC: ASME B18.22.1 MATERIAL: ZINC PLATED QTY: 16 EA	9B 5330	9		16/EA	3.3
506	M-1	2	M	PUMP, GEAR, ITEM NAME: GEAR, PUMP, 10" FLANGE PORTS 1000 GPM P/N: RS324A MATERIAL: CAST IRON OEM: VIKING PUMP	9B 4320	9		1/EA	3.3
507	M-2	2	M	COUPLING, SHAFT, FLOATING SHAFT COUPLING, P/N: SX278-6 MATERIAL: VARIOUS RMKS: SIM TO LOVEJOY TYPE SX (LOV SX278-6 CP 3-7/16" X 3- 7/16" - 7/8KWBE-BSE=71" -SCK)	9B 4730	9		1/EA	3.3
508	P-1	2	M	PIPE, 8" PIPE, 8.625" OD X .322 THK, SCH 40 SPECS: ASTM A53 GR B, ANSI B36.10 SEAMLESS MATERIAL: CARBON STEEL	9B 4710	9		10/FT	3.3
509	P-2	2	M	PIPE, 6" PIPE, 6.625" OD X .280 THK, SCH 40 SPECS: ASTM A53 GR B, ANSI B36.10 SEAMLESS MATERIAL: CARBON STEEL	9B 4710	9		1/FT	3.3
510	P-3	2	M	TUBE, 1/4" OD TUBE X .035 THK SPECS: ASTM A269, MATERIAL: STAINLESS STEEL TYPE 304	9B 4710	9		20/FT	3.3
521	F-8	2	M	EXPANSION JOINT, 10" X 8", CONC EXP, JOINT, 150# 6" F/F, BUTYL/NYLON, SIM TO FLEXICRAFT ASA B16.5 DRILLING	9B 4730	9		2/EA	3.3

5/15

Ship/HULL NO.	Ship/HULL NAME	JON:	ICN: 11 89NSC85004R	Rev:	Key Op: A01
YON-305	FLCY SASEBO FUEL DEPT				

TITLE:

SA YON-0120D for Cargo Pump; accomplish

MATERIAL LIST									
ITEM NO.	PC NO.	REF NO.	M / C	DESCRIPTION	STOCK NUMBER	HAZ MAT	DOC NO.	QTY REQD/ UNIT	PARA NO.
524	F-11	2	M	BOLT, HX HD,BOLT, HEX HD, 3/4-10UNC-2A - 2-1/2 LG MIL SPEC: MIL-DTL-1222 GR5 MATERIAL: STEEL	9B 5330	9		88/EA	3.3
525	F-12	2	M	NUT,NUT, 3/4-10UNC-2B MIL SPEC: MIL-DTL-1222 GR5 MATERIAL: STEEL	9B 5330	9		88/EA	3.3
526	F-13	2	M	STUD BOLT,STUD BOLT, HEX HD, 7/8-9UNC-2A - 3-1/2 LG MIL SPEC: MIL-DTL-1222 GR5 MATERIAL: STEEL	9B 5330	9		24/EA	3.3
527	F-14	2	M	NUT,NUT 7/8-9UNC-2B MIL SPEC: MIL-DTL-1222 GR5 MATERIAL: STEEL	9B 5330	9		24/EA	3.3
528	F-15	2	M	TEE, FLANGE,8", TEE,FLANGED 150#, ASTM A536, DUCTILE IRON, SIM TO SMITH COOPER P/N 38T 1080	9B 4730	9		1/EA	3.3
529	F-16	2	M	STRAINER,8" SIMPLEX STRAINER, 125#,FLGD, ASTM A126 CL "B", CAST IRON, SIM TO WATTS SERIES 97FB-CIB, ANSI B16.1, CLASS 125#	9B 4730	9		1/EA	3.3
533	F-20	2	M	PLATE,PHOTOGRAPHIC,PLATE,P HOTOGRAPHIC	9B 6750 014336627	9		1/EA	3.3
538	F25 D	2	M	VALVE,LINEAR,DIRECT,VALVE,LI NEAR,DIRECT	9B 4820 012068957	9		2/EA	3.3
539	F-26	2	M	ROD, METAL,1" ROD, ASTM -A36, STEEL	9B 9530	9		10/FT	3.3

6/15

Ship/HULL NO. YON-305	Ship/HULL NAME FLCY SASEBO FUEL DEPT	JON:	ICN: 89NSC85001R	Rev:	Key Op: A01
--------------------------	---	------	------------------	------	----------------

TITLE:

SA YON-0120D for Cargo Pump; accomplish

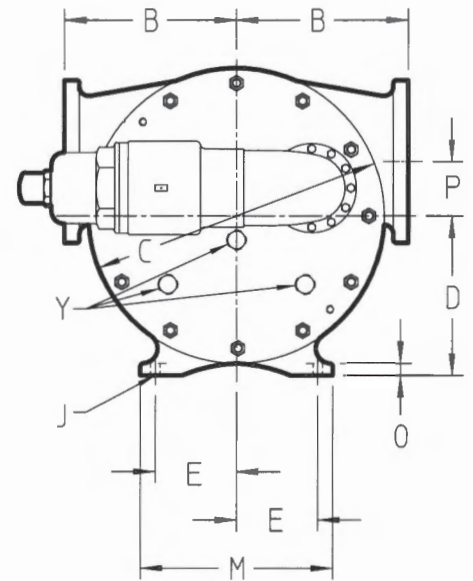
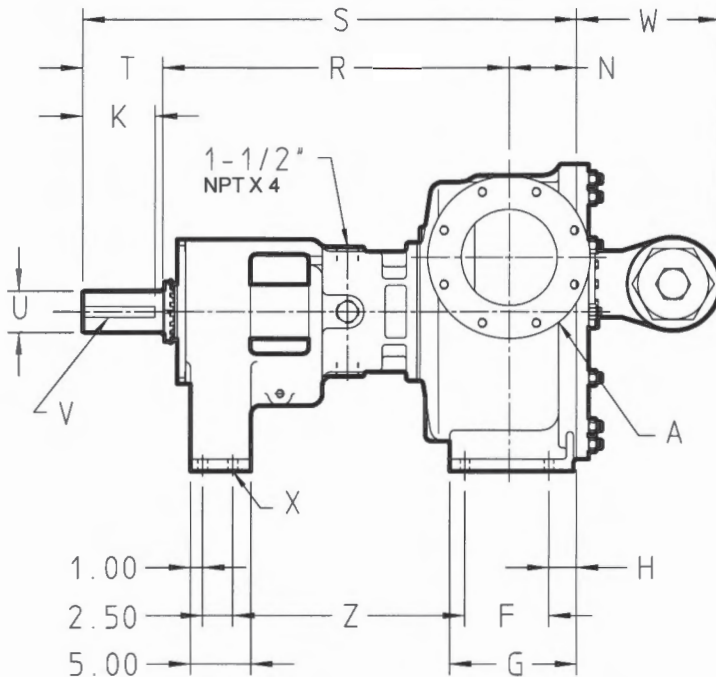
DATA RECORD					
OPERATING PARAMETER : ACCEPTANCE CRITERIA (*Data is for information purposes only)		TIME OF READING			
		START	LAST		
		TIME	:	TIME	:
ENGINE	Speed (RPM)		RPM		RPM
	Oil Pressure (PSIG)		PSIG		PSIG
	Water Temperature (°F)		°F		°F
	PTO Case Temperature (°C)		°C		°C
PUMP	Discharge Pressure (PSIG)		PSIG		PSIG
	Suction Pressure (PSIG)		PSIG		PSIG
	Flow (GPM)		GPM		GPM
	Mechanical Seal Temperature (°C)		°C		°C
	Pump Case Temperature (°C)		°C		°C
	Mechanical Seal Leakage : NONE	RESULT: ☐ SAT / ☐ UNSAT			
	Disturbed Joint Leakage : NONE	RESULT: ☐ SAT / ☐ UNSAT			
COMPLETION OF THE TEST					
RESPONSIBILITY		PRINT NAME, SIGN & DATE			
LWC					
FLCY SASEBO					

UNIVERSAL PRODUCT LINE: CAST IRON — JACKETED PUMPS

224A Series™, 1224A Series™, 1224A-BXB Series™, 4224A Series™, 4224B Series™, 324A Series™, 1324A Series™, 4324A Series™

Section	1402
Page	1402.13
Issue	I

DIMENSIONS – N, R & RS SIZES – JACKETED BRACKET (324A, 4324A)



Model Number			A (in)		B	C	D	E	F	G	H	J	K	M
Packed	O-Pro® Seal	Stuffing Box Seal												
N324A	N1324A	N4324A	② 6	in	9.75	17.25	9.50	5.00	6.25	8.69	1.62	0.69	4.50	12.00
				mm	248	438	241	127	159	221	41	18	114	305
R324A	R1324A	R4324A	② 8	in	14.25	24.50	13.25	6.75	7.00	10.56	2.31	0.78	6.00	16.00
				mm	362	622	337	171	178	268	59	20	152	406
RS324A	RS1324A	RS4324A	② 10	in	14.25	24.50	13.25	6.75	7.00	13.12	4.81	0.88	6.00	16.46
				mm	362	622	337	171	178	333	122	22	152	418

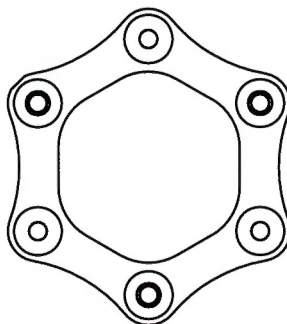
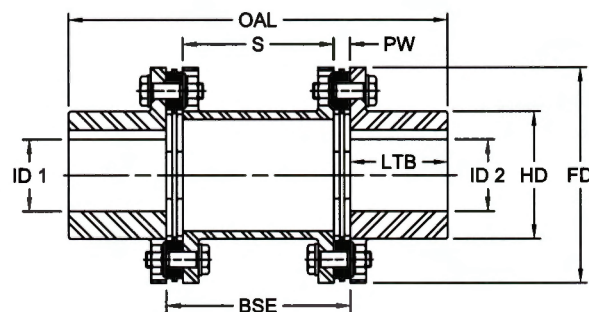
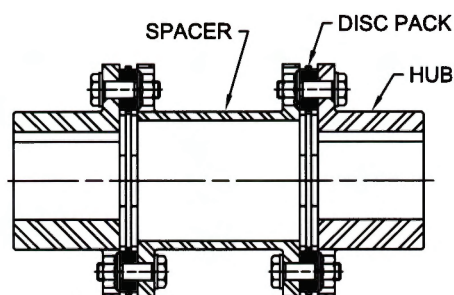
Model Number				N	O	P	R	S	T	U (in)	V (in)	W	X	Y (in)	Z
Packed	O-Pro® Seal	Stuffing Box Seal													
N324A	N1324A	N4324A	in	4.50	1.00	3.00	26.00	36.50	6.00	2.44	.62 x .31	8.63	0.69	N/A	18.94
			mm	114	25	76	660	927	152			219	18		481
R324A	R1324A	R4324A	in	5.62	1.00	4.50	28.75	41.00	6.62	3.44	.88 x .44	12.00	0.69	1.25 NPT	19.25
			mm	143	25	114	730	1041	168			305	18		489
RS324A	RS1324A	RS4324A	in	8.12	1.30	4.50	28.55	43.49	6.62	3.44	.88 x .44	12.00	0.88	1.25 NPT	19.25
			mm	206	33	114	725	1105	168			305	22		489

NOTE: The N size is standard with a jacketed bracket and non-jacketed head and non-jacketed relief valve, while the "R" size is standard with a jacketed bracket, a jacketed head, and a non-jacketed relief valve. "RS" contact factory for jacketing options.

② Ports are suitable for use with Class 125 ANSI cast iron companion flanges or flanged fittings.

ENCLOSURE (B) - 1/1

8/15



SX-6 Dimensional Data

Size	OAL		S		PW		LTB		ID1-ID2 Max Bore ³		BSE ⁷ Standard		FD		HD	
	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm	in	mm
90-6	5.50	140	1.77	45	0.295	7.5	1.57	40	1.50	38	2.36	60	3.54	90	2.28	58
110-6	7.44	189	2.84	72	0.331	8.4	1.97	50	1.81	46	3.50	89	4.33	110	2.56	65
132-6	8.97	228	3.59	91	0.331	8.4	2.36	60	2.38	60	4.25	108	5.20	132	3.31	84
158-6	10.40	264	4.00	102	0.441	11.2	2.76	70	2.76	70	4.88	124	6.22	158	3.86	98
185-6	11.80	300	4.40	112	0.551	14.0	3.15	80	3.13	80	5.50	140	7.28	185	4.41	112
202-6	13.33	339	5.03	128	0.610	15.5	3.54	90	3.50	90	6.25	159	7.95	202	4.92	125
228-6	14.76	375	5.50	140	0.689	17.5	3.94	100	3.94	100	6.88	175	8.98	228	5.51	140
255-6	16.81	427	6.14	156	0.807	20.5	4.53	115	4.31	110	7.75	197	10.04	255	6.10	155
278-6	18.47	469	6.96	177	0.835	21.2	4.92	125	4.88	124	8.63	219	10.95	278	6.85	174
302-6	19.89	505	7.33	186	0.961	24.4	5.32	135	5.31	135	9.25	235	11.89	302	7.48	190
325-6	21.42	544	7.95	202	1.024	26.0	5.71	145	5.75	145	10.00	254	12.80	325	8.07	205
345-6	22.83	580	8.41	214	1.110	28.2	6.10	155	6.06	155	10.63	270	13.58	345	8.55	217
380-6	25.08	637	9.16	233	1.260	32.0	6.70	170	6.69	170	11.68	297	14.96	380	9.37	238
410-6	27.18	690	10.01	254	1.307	33.2	7.28	185	7.13	180	12.62	321	16.14	410	10.04	255
440-6	28.55	725	10.32	262	1.433	36.4	7.68	195	7.69	195	13.19	335	17.32	440	10.75	273

Notes: ■ 3 indicates: The maximum bores shown are for cylindrical or taper shafts with keys. For splines and other type bores contact Lovejoy Technical Support.

■ 7 indicates: BSE is the distance between shaft ends and is a variable parameter.

■ Metric dimensions are rounded to the nearest 'mm' except where otherwise shown.



< IC 189N3C8500 / A04 Disc Couplings SX Spacer Style (6-Bolt) Installation Guide

1.0 INTRODUCTION:

The following document is intended for the explicit use of Lovejoy customers to aid in the installation of Lovejoy power transmission products. The information may be considered privileged and should only be disseminated as an active part of conducting business with Lovejoy, Inc.

Although the coupling may have been properly specified during the design and selection process before the coupling was ordered, operational conditions could possibly have changed prior to installation. Lovejoy, Inc. provides the information and technical support necessary to ensure the appropriate coupling selection was made relative to the product specifications and limitations of Lovejoy's power transmission products. The end user is ultimately responsible for verifying the suitability of the final coupling selection based on the actual service conditions at the time the coupling is installed.

Correct installation and alignment practices will ensure longer coupling life, trouble free operation, and a safer operating environment for the coupling. Please thoroughly review all of the instructions in this document prior to installing this coupling and placing it in operation. Proper safety guidelines and practices should always be followed during every phase of the installation.

This installation document is considered part of the purchased product and should be retained for future reference.

2.0 SAFETY:

Accidents involving rotating equipment may result in loss of life, serious bodily harm, or property damage. The purchaser of this equipment must assure that the equipment is properly assembled, installed, safeguarded, operated, and maintained. This equipment should never be operated at, or subjected to, conditions that exceed manufacturer's specifications.

Consult all applicable Federal, State and local laws and regulations covering the safe operation and maintenance of equipment, including, without limitation, the USDOL-OSHA "Lockout / Tag-out" procedure set forth in 29 CFR 1910.147.

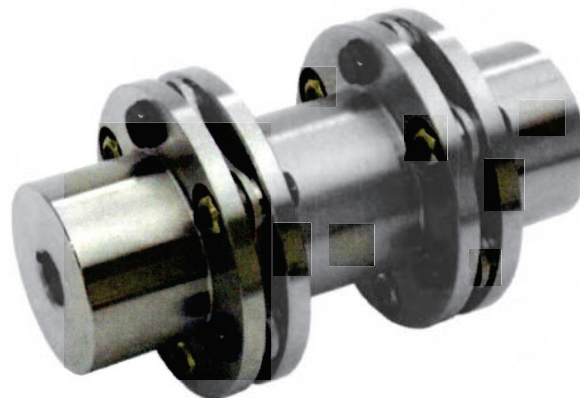
Because of the possible danger to persons or property from accidents which may result from the improper use or unapproved modifications of the product, this product must be installed, maintained and operated in accordance with the procedures, standards, and engineering specifications specified in the product literature. To assure safe operation, this product should be inspected in accordance with the instructions described in this document. Proper guards and any suitable safety equipment or procedures as may be necessary, or as may be specified in safety codes, should be installed by the user. Safety equipment, coupling guards, and shields are not provided by, nor are they the responsibility of Lovejoy, Inc.

Symbols and text format used in this document may contain safety information and will appear similar to the following:



Warning! This symbol indicates safety measures which must be observed to avoid personal injury.

CAUTION! This symbol indicates safety measures which must be observed to avoid damage to coupling.



3.0 PRODUCT INSPECTION:

Prior to installation, the coupling should be examined for signs of damage that may have occurred as a result of shipping or handling. Refer to the following chart (Table-1) to ensure all the ordered parts were included with the shipment.

Note: Standard SX Disc Couplings are shipped unassembled.

Table 1 - Components

Size	Hubs	Spacer	Disc Pack Kit		
			Disc Pack	Bolts	Lock Nuts
SX 090-6	2	1	2	12	12
SX 110-6	2	1	2	12	12
SX 132-6	2	1	2	12	12
SX 158-6	2	1	2	12	12
SX 185-6	2	1	2	12	12
SX 202-6	2	1	2	12	12
SX 228-6	2	1	2	12	12
SX 255-6	2	1	2	12	12
SX 278-6	2	1	2	12	12
SX 302-6	2	1	2	12	12
SX 325-6	2	1	2	12	12
SX 345-6	2	1	2	12	12
SX 380-6	2	1	2	12	12
SX 410-6	2	1	2	12	12
SX 440-6	2	1	2	12	12

For maximum protection, the coupling and all components should be stored in the original packaging. All parts should be measured prior to installation to ensure correctness of parts to meet the application requirements; such as the hub bore diameter, shaft diameter, shaft separation, bolt lengths, key sizes, etc.



Warning! Before performing the coupling installation, make sure the machinery is made safe. Disconnect and lock out all power to the equipment. No part of the installation should be performed on moving, non secure, or unstable equipment.

CAUTION! Lovejoy manufactured the coupling interface based on the equipment and shaft data supplied by the purchaser. Lovejoy is not responsible for inaccurate or incomplete information supplied by the purchaser.

It is the purchasers' responsibility to assure that the interface connections between the coupling and the connected equipment are capable of handling the anticipated loads.

4.0 REQUIRED TOOLS:

- Vernier Calipers
- Calibrated Torque Wrench
- Sockets and appropriate open end wrenches
- Alignment Equipment
- Appropriate hoist or lifting equipment

5.0 COUPLING AND COMPONENT PREPARATION:

5.1 Prior to installation, all exposed surfaces of the coupling and components, including hubs, disc packs, and hardware should be thoroughly cleaned to remove any protective coatings normally applied by Lovejoy at the factory. These coatings are applied as corrosion protection for the coupling surfaces during shipping. All coupling parts, equipment components, shafts, and keyways must be clean and free of any foreign materials prior to attempting assembly or installation. A clean cloth dampened with a nonflammable solvent should be sufficient for this cleaning.

5.2 All sleeves, seals, hub bores, shafts, keys, and keyways should be checked for raised metal, nicks, burrs, dents, gouges, etc., and dressed or repaired accordingly prior to installation.

5.2.1 Prior to removing any existing coupling, establish and record the Distance Between Shaft Ends (BSE) between the driver and driven shafts. Compare this value with the BSE dimension for the Disc Coupling as purchased.

5.2.2 Once all necessary measurements have been taken and all components are confirmed to be correct, remove any existing coupling and dress the shafts of the driver and driven equipment.

5.2.3 If the final shaft separation is the same as the specified BSE for the Lovejoy Disc coupling, then the hubs can be mounted flush with the ends of the driver and driven shafts.

5.2.4 If the final shaft separation is different than the specified BSE of the Lovejoy Disc coupling, then the hubs must be mounted on the driver and driven shafts so that the distance between the hub flanges match the 'BSE' dimension of the coupling as purchased. If this dimension cannot be met, please contact your coupling supplier.

CAUTION! *The amount of hub engagement on the shaft should be 80% of the shaft diameter or greater for an interference fit.*

5.3 Lovejoy machines bores in all Lovejoy disc coupling hubs with 'inch' dimensioned straight bores and keyways to meet the industry accepted **ANSI/AGMA 9002-B04** Standards' tolerance for an interference, or shrink fit bore unless otherwise specified. Tapered and spline bores typically require special consideration.

5.4 Lovejoy machines bores in all Lovejoy disc coupling hubs with 'metric' dimensioned straight bores and keyways to meet the industry accepted **ANSI/AGMA 9112-A04** Standards' tolerance for an interference, or shrink fit bore unless otherwise specified. Tapered and spline bores typically require special consideration.

5.5 For all Lovejoy disc coupling hubs with taper bores and taper bores with keyways, Lovejoy manufactures these hubs with bores using tolerances and specifications as supplied by the customer. Taper bores will be tested with plug gauges usually supplied by the customer or included in the cost of the coupling.

6.0 HUB INSTALLATION (INTERFERENCE FIT):

6.1 Prior to installing any coupling, it is important to make sure the equipment where the coupling is being installed is made safe and no part of this installation will be done on non-secured, unsafe, or moving equipment. Ideally, the equipment should be disconnected from any power source using the Lockout/Tag-out procedures defined by OSHA. Consult with local, State, and Federal laws and regulations covering safe operation and maintenance of equipment.



Warning!

When installing hubs, consult with all applicable Federal, State, and local laws and regulations covering the safe operation and maintenance of equipment, including, without limitation, the USDOL-OSHA "Lockout/Tag-out" procedure set forth in 29 CFR 1910.147.

6.2 Lovejoy disc coupling hubs are shipped with Interference Fit bores unless otherwise specified. Prior to installing the coupling, measure the shaft diameters and the hub bores to ensure proper fits. With interference fits, the hub bore diameters should be slightly less than the shaft diameters and the hubs will need to be heated prior to mounting on the shafts.

Note, the Engineering Data section of the *Lovejoy Power Transmission Products Catalog* on-line contains the actual bore and keyway sizes supplied by Lovejoy for specific shaft sizes. Please visit the Lovejoy website at <http://www.lovejoy-inc.com>.

6.3 Install the keys in the shaft keyways. Each key should fit snugly in the keyway with no side to side movement. The key should be the same length as the hub to maintain dynamic balance and transmit the maximum torque.

6.4 When mounting hubs that have been machined with an interference fit, there should be a slight clearance between the top of the key and the keyway. Without this clearance, when the hub cools, the keyway will rest on the key creating high stresses in the hub that could cause the hub to fail.

6.5 Heat the hubs and slide them on the shafts with flange end located flush with the end of the shaft. Make sure the hub is heated uniformly to a temperature of at least 350° F (177° C). Steps 6.5.1 through 6.5.4 provide procedures that can be used when heating interference fit hubs.

6.5.1 Oil Bath Heating is usually limited to 350° F. (177° C), or some temperature that is less than the flash point of the oil used. Special handling devices are required to support the hub in the oil bath such as tongs, threaded rods or eye-bolts placed in puller holes, etc. The hubs should not rest on the bottom of the oil bath container and must remain in place for a period of time ample to heat the hub all the way through.



Warning!

If an oil bath is used, the couplings will need to be heated to approximately 350° F (177° C) or more, so the oil must have a flash point above 350° F (177° C).

6.5.2 Induction Heating can be used as long as the temperature rise in the hub is uniform and controlled.

Lovejoy, Inc.
World Headquarters
 2655 Wisconsin Avenue
 Downers Grove, IL 60515
 630-852-0500
 630-852-2120 Fax
info@lovejoy-inc.com

where the world turns for

ENCLOSURE (D) - 2/6
 Page 2 of 6

www.lovejoy-inc.com

ISO 9001:2008 Certified

Version Date: 25-January-2013

11/15

6.5.3 Oven Heating offers definite advantages over oil bath heating. Parts can be heated to higher temperatures, usually 450° F to 600° F (235° C to 315° C). 600° F is the maximum temperature where the steel hub does not go through an annealing process and yet can still be handled with heat resistant gloves. When heating the hubs in an oven, place them on a rack and do not rest the hubs on the oven surface. The hubs should remain in the oven for a sufficient period of time to heat the hub all the way through.

6.5.4 Open Flame Heating is typically not recommended. If the hub is being heated with an oxyacetylene, or blow torch, use an excess acetylene mixture. Mark the hub body at the top, center, and bottom along the length of the hub with heat resistant crayons, one with a 350° F (177° C) melt temperature and another with a 450° F (232° C) melt temperature. The hub should be sitting elevated on refractory bricks oriented to allow the flame to flow through the hub. With a "Blue Flame" or "Rosebud" torch, direct the flame towards the hub bore using constant motion to avoid overheating any single area. Once the heat sensitive crayon marks melt, the hub should be ready for mounting.



Warning! Do not use an open flame in a combustible atmosphere or near combustible materials.

CAUTION!

Do not "spot" heat the hub in single areas or distortion of the hub could occur.

CAUTION!

Do not exceed 600° F (315° C) during the heating process. Excessive heat can soften, or anneal the hub, reducing the strength of the steel thus affecting the performance characteristics of the hub.

CAUTION!

Use extreme care when handling heated hubs to avoid injury to personnel.

6.6 If either hub was manufactured with a clearance, or slip fit bore, slide these hubs onto the appropriate shafts. Align the flanged end of the hub flush with the end of the shaft and key. These hubs will be held in place with set screws.



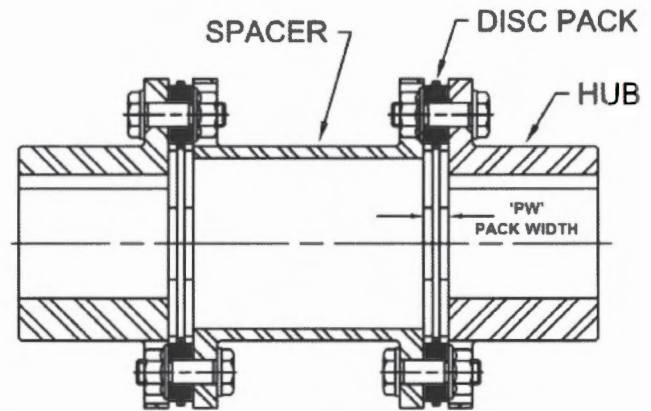
Warning! Always allow the hubs to cool to ambient (room) temperature prior to continuing with the installation.

7.0 INSTALL SPACER AND DISC PACKS:

7.1 Always allow the hubs to cool to the ambient, or room temperature prior to proceeding with the installation.

7.2 If the equipment is not already in the operating position, move the equipment into position ensuring the measured distance between the shaft ends matches the "BSE" dimension of the coupling as ordered from the supplier. The shafts should be aligned using an acceptable optical method, preferably laser alignment. When the equipment shafts are aligned and the gap between the shaft ends match the BSE of the coupling as ordered, tighten the equipment anchor bolts to the manufacturer's specified torque.

7.3 Apply a light film of grease to both sides of all the bushings on both disc packs to prevent the bushings from binding and twisting when the lock nuts are tightened. This could cause distortion in the disc packs that could lead to premature disc pack failure.



7.4 Carefully move the spacer into position between the hubs without allowing the spacer to impact against either hub. Always maintain support for the spacer until all bolts are in place.

7.5 Slide one disc pack between the flanges of the hub and the spacer. Insert the bolts through the small holes in the hub flange and through the bushings in the disc pack. The lock nuts will be placed on the bolts through the larger holes in the spacer. Hand tighten the lock nuts against the disc pack bushings.

7.6 Insert the bolts through the small holes in the spacer flange and through the bushings in the disc pack. The lock nuts will be placed on the bolts through the larger holes in the hub. Hand tighten the lock nuts against the disc pack bushings.

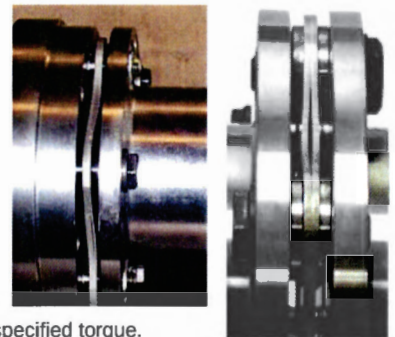
7.7 Use an open end or box end wrench to prevent the bolts from turning, then with a calibrated torque wrench, tighten the lock nuts on this disc pack using the industry standard procedure for tightening the nuts first to 50%, then 75%, then to the final torque specified in Table-2. Tighten all of the lock nuts on one side of the disc pack before tightening the lock nuts on the second side.

7.8 Repeat steps 7.5 through 7.7 to install the disc pack in the other end of the coupling. If the space between the flange on the second hub and the spacer measures less than the standard PW Width from Table-3, or is too narrow for the disc pack to fit between the flanges, the equipment may need to be repositioned to install this disc pack.

7.9 Measure the gaps between the hubs and spacer on both ends of the coupling. The gaps should match the "PW", or Pack Width dimension specified in Table-3.

7.10 Inspect the disc packs to ensure there is no waviness or gaps between the layers in the disc pack. If the gaps between the hubs and spacer flange do not match the "PW" dimension, or pack width specified in Table-3, the disc pack could take on a wavy appearance.

To correct this, some axial adjustment of the equipment may need to be performed. If any of the disc pack bushings rotated while tightening the lock nuts, the layers, or leaves in the unitized disc pack could show signs of bowing, or separating. This can be corrected by loosening the lock nuts, then add a slight film of grease between the bushing and lock nut, then retighten the lock nuts to the specified torque.



Lovejoy, Inc.
World Headquarters
2655 Wisconsin Avenue
Downers Grove, IL 60515
630-852-0500
630-852-2120 Fax
info@lovejoy-inc.com

where the world turns for
Lovejoy
ENCLOSURE (D) - 3/6
Page 3 of 6

www.lovejoy-inc.com

ISO 9001:2008 Certified

Version Date: 25-January-2013

12/15

7.0 INSTALL SPACER AND DISC PACKS (CONT'D):

7.11 Measure the gap between the hub and spacer on the second end of the coupling. This gap should fall within the range specified as 'PW' High and Low in Table-3.

7.12 Check both angular and axial alignment at each disc pack location. Measure the gap between the hub and spacer flanges at four or more locations around the coupling at each disc pack, preferably the 3:00, 6:00, 9:00, and 12:00 positions. Compare these measurements with the 'PW' Disc Pack Width value specified in Table-3. All measurements must fall within the PW/Low and PW/High values. If the gap for the disc pack width falls outside this high/low range, make adjustments in the equipment alignment before continuing.

7.12 Remove any tooling and material away from the coupling and equipment. Install the coupling guard per OSHA requirements and remove the Lockout / Tagout kit from the power supply. The equipment can then be started up and tested. The coupling and equipment should run smoothly. If vibration is detected it could indicate there is an issue with alignment or other problems possibly related to the motor, coupling, or driven equipment. These issues should be resolved prior to placing this coupling into operation.

8.0 STANDARD COUPLING INSPECTION AND MAINTENANCE:

8.1 A static coupling inspection can be performed with the equipment stopped and made safe per OSHA, or other federal, state, and local government requirements. The coupling should be inspected for any signs of damage, wear, or fatigue that may have occurred during normal operation of the equipment. The coupling should be rotated to facilitate a full 360° inspection. The following steps should be taken:

8.1.1 Inspect all fasteners for signs of damage, wear, or fatigue and make sure all the bolts (and lock nuts) are tightened to the torque specified in Table-2. Replace and retighten the bolts as necessary.

8.1.2 Inspect the disc packs for cracks, separation of the layers or leaves within the disc pack (see Section 7.10) and any other abnormalities that may be evidence of fatigue or wear. The disc packs, bolts, and locknuts should be replaced if necessary.

8.1.3 Inspect the disc pack bushings (6 or 8 per disc pack) for signs of any cracks or breaks and replace the complete disc pack if necessary. Replace the bolts and locknuts anytime replacement disc packs are installed. Replacement disc packs kits come with hardware.

8.1.4 Inspect the hubs and spacer for distortion or signs of damage and wear. If necessary, replace these components.

8.2 Dynamic Coupling Inspection can be performed while the coupling is in operation. The disc packs can be inspected through a protective shield or mesh screen using a strobe light. Use the following steps when performing a dynamic inspection.

8.2.1 If there is vibration present in the system, the cause may not necessarily come from the coupling. There could be balance issues with the driving or driven equipment which could cause damage to the equipment or coupling if left unresolved.

8.2.2 Provide access to viewing each disc pack through a safety shield or mesh screen so that the disc packs can be viewed using a strobe light with the coupling operating under load.

8.2.3 Adjust the strobe light so that each disc pack appears to be stationary, or rotating very slowly. Adjust the strobe on intervals to ensure the inspection can be performed on all the disc pack bushings and disc segments.

8.2.4 Check for cracks, breaks, or looseness in the bushings.

8.2.5 Check each disc pack for any bowing or separation of the layers or leaves in the disc pack (see section 7.10).

8.2.6 Check each disc pack for waviness or distortion (see section 7.10) that could indicate misalignment or improper 'PW' spacing.

8.2.7 Check both disc packs for any broken, frayed, or cracked layers or leaves. Specifically look for broken leaves on the outer layers of each disc pack next to the bushings. This is the symptom for the early stages of a normal disc fatigue failure caused by misalignment.

8.3 If any of the conditions specified in sections 8.1 and 8.2 exist, the cause should be resolved prior to placing the equipment back in operation. If the equipment needs to be moved or disc packs replaced, always recheck the equipment alignment before operating.

9.0 DISC PACK REPLACEMENT:

9.1 Remove the disc pack by first removing the lock nuts and bolts that hold the disc pack in place. Discard these fasteners and use the new fasteners supplied with the replacement disc pack kit when the new disc pack is installed. If the disc pack does not drop out from between the hub and spacer flanges, one of the pieces of equipment may need to be moved.

9.2 Position the replacement disc pack between the hub and spacer flanges. Insert the disc pack bolts through the small holes in the hub flange, then through the bushings in the disc pack.

9.3 Place the lock nuts on the bolts through the large holes in the spacer flange and tighten until snug against the disc pack bushing. If one of the pieces of equipment needs to be moved back into place, do not tighten the lock nuts with the torque wrench until the equipment is in position and aligned.

9.4 Install the remaining disc pack bolts through the flange on the spacer. Hand tighten the lock nuts through the clearance holes in the hub until they contact the disc pack bushings.

9.5 Tighten the lock nuts on the disc pack bolts. Use an open or box end wrench to hold the bolt heads while tightening the nuts with a calibrated torque wrench. The socket will need to fit inside the clearance holes in the flanges. Tighten the lock nuts using the industry standard procedure of first tightening the nuts to 50%, then 75%, then the full amount of the torque specified in Table-2. Tighten all the nuts on one side of the disc pack prior to tightening the second side.

9.6 Repeat steps 9.1 through 9.5 for the second disc pack.

9.7 If equipment was moved, check alignment first using the optical, or laser method, then measure the disc pack gap width at four or more locations around the coupling, preferably at the 3:00, 6:00, 9:00, and 12:00 positions and compare these values with the 'PW/Low' value and the 'PW/High' value in Table-3. If any of the measurements fall outside the high/low range, make adjustments in the equipment alignment before continuing.

9.8 Remove any tooling and material away from the shafting and coupling. Install the coupling guard per OSHA requirements and remove the Lockout / Tagout kit from the power supply. The equipment can then be started up and tested. The coupling and

equipment should run smoothly. If vibration is detected it could indicate there is an issue with alignment or other problems possibly related to the motor, coupling, or driven equipment. These issues should be resolved prior to placing this coupling into operation.

Table 2 - Lovejoy Disc Pack Bolts & Locknuts

Size	Socket Size		Torque Wrench	Clearance Hole		Bolt Torque			
	Bolt Size mm	Nut Size mm		Diameter in	Diameter mm	Lubricated Threads		Dry Threads	
			ft-lbs			Nm	ft-lbs	Nm	
SX 90-6	10	10	1/4	0.63	16	8	11	10	14
SX 110-6	13	13	3/8	0.827	21	18	24	22	30
SX 132-6	13	13	3/8	0.827	21	18	24	22	30
SX 158-6	17	15	3/8	1.024	26	35	48	44	60
SX 185-6	19	18	3/8	1.26	32	59	80	74	100
SX 202-6	22	21	1/2	1.378	35	89	120	111	150
SX 228-6	24	24	1/2	1.457	37	136	184	170	230
SX 255-6	30	30	3/4	1.772	45	266	360	332	450
SX 278-6	30	30	3/4	1.811	46	266	360	332	450
SX 302-6	32	32	3/4	1.968	50	354	480	443	600
SX 325-6	32	32	3/4	2.047	52	354	480	443	600
SX 345-6	36	36	3/4	2.205	56	460	624	575	780
SX 380-6	41	41	1	2.362	60	649	880	811	1100
SX 410-6	46	46	1	2.677	68	885	1200	1106	1500
SX 440-6	50	50	1	2.913	74	1180	1600	1475	2000

Note: Please be advised that the Clearance Hole Diameter of the coupling may be smaller than the Outside Diameter of some typical size socket. The socket's OD may have to be turned down to 0.76 mm (0.030") smaller than the coupling's clearance hole diameter.

Table 3 - Allowable Misalignment

Size	Disk Pack Width (PW) - Gap								Misalignment				
	Standard Pack Width (PW)				PW + / - Dimensions				Axial ±ΔKa		Angular Max Degree	Parallel Max inch mm	
	PW		Tolerance + / -		Low		High						
	in	mm	in	mm	in	mm	in	mm	in	mm			
SX 90-6	0.295	7.5	0.003	0.08	0.292	7.43	0.298	7.58	0.030	0.75	1.5°	Spacer Length x 0.025"	Spacer Length x 0.6mm
SX 110-6	0.331	8.4	0.004	0.11	0.327	8.30	0.335	8.51	0.039	1.00			
SX 132-6	0.331	8.4	0.005	0.13	0.326	8.27	0.336	8.53	0.051	1.30			
SX 158-6	0.441	11.2	0.006	0.16	0.435	11.05	0.447	11.36	0.059	1.50			
SX 185-6	0.551	14.0	0.007	0.19	0.544	13.82	0.558	14.19	0.071	1.80			
SX 202-6	0.610	15.5	0.007	0.19	0.603	15.31	0.617	15.69	0.075	1.90	1°	Spacer Length x 0.017"	Spacer Length x 0.4mm
SX 228-6	0.689	17.5	0.008	0.21	0.681	17.29	0.697	17.71	0.083	2.10			
SX 255-6	0.807	20.5	0.009	0.24	0.798	20.27	0.816	20.74	0.091	2.30			
SX 278-6	0.835	21.2	0.010	0.26	0.825	20.94	0.845	21.46	0.102	2.60			
SX 302-6	0.961	24.4	0.011	0.29	0.950	24.12	0.972	24.69	0.110	2.80			
SX 325-6	1.024	26.0	0.013	0.33	1.011	25.68	1.037	26.33	0.126	3.20			
SX 345-6	1.110	28.2	0.014	0.35	1.096	27.86	1.124	28.55	0.134	3.40			
SX 380-6	1.260	32.0	0.015	0.38	1.245	31.62	1.275	32.38	0.150	3.80			
SX 410-6	1.307	33.2	0.016	0.41	1.291	32.79	1.323	33.61	0.161	4.10			
SX 440-6	1.433	36.4	0.017	0.44	1.416	35.96	1.450	36.84	0.173	4.40			

Lovejoy, Inc.
World Headquarters
 2655 Wisconsin Avenue
 Downers Grove, IL 60515
 630-852-0500
 630-852-2120 Fax
 info@lovejoy-inc.com


ENCLOSURE (D) - 5/6
 Page 5 of 6

www.lovejoy-inc.com

ISO 9001:2008 Certified

Version Date: 25-January-2013

14/15

Table 4 - Maximum Radial Load

Disc Pack Size	Maximum Radial Load per Disc Pack ¹	
	lbf (Pound-Force) ²	N (Newton) ²
6 - Bolt		
90-6	90	420
110-6	99	440
132-6	157	700
158-6	225	1 000
185-6	337	1 500
202-6	427	1 900
228-6	585	2 600
255-6	764	3 400
278-6	967	4 300
302-6	1,236	5 500
325-6	1,439	6 400
345-6	1,686	7 500
380-6	2,113	9 400
410-6	2,630	11 700
440-6	3,240	14 500

Disc Pack Size	Maximum Radial Load per Disc Pack ¹	
	lbf (Pound-Force) ²	N (Newton) ²
8 - Bolt		
278-8	1,281	5 700
302-8	1,754	7 800
325-8	2,001	8 900
345-8	2,360	10 500
380-8	2,945	13 100
410-8	3,709	16 500
440-8	4,429	19 700
475-8	5,171	23 000
505-8	6,182	27 500

- Notes:**
1. Values are based on the shear force calculated for the minor diameter of the disc pack bolts.
 2. The radial load capacity for stainless steel fasteners is approximately 50% of the values listed for standard hardware.